

National Cybersecurity Law



Cybersecurity Law & Policy Project | Prepared
by Vedant Bandal and Team





Introduction & Need for the Law

- Purpose: Establish a comprehensive cybersecurity framework for VIT's digital ecosystem
- Importance: Protect digital assets, academic data, and student privacy
- Current Threats: Rising cyberattacks targeting educational institutions globally
- Expected Outcomes: Enhanced security posture, institutional trust, and regulatory compliance





Key Definitions

Cybercrime: Illegal activity using computers or networks. Hacker: Person who exploits system vulnerabilities. Ethical Hacker: Authorized expert who tests security. Malware: Malicious software designed to damage systems. Ransomware: Malware that encrypts data for ransom. Phishing: Deceptive attempt to steal sensitive information. Data Breach: Unauthorized access to confidential data. Personal Data: Identifiable information about an individual. Encryption: Encoding data to prevent unauthorized access. Digital Signature: Cryptographic authentication of digital documents.

Cybercrime

Encryption

Data Breach

Phishing



Citizens' Digital Rights

The National Cybersecurity Law for VIT's – Cyberland guarantees fundamental digital rights to every citizen, ensuring privacy, data control, and access to justice in the digital realm.

Right to Privacy & Secure Data

Right to Data Deletion & Information Access

Right to Report Cybercrime & Fair Investigation





Cyber Offenses & Penalties

The law defines strict penalties for all cyber offenses. Unauthorized Access & Data Theft: up to 5 years imprisonment. Identity Theft & Phishing: fines + 3–7 years. Ransomware & Malware Distribution: up to 10 years. Cyberbullying: counseling + fines. Financial Fraud: asset seizure + imprisonment. Cyber Terrorism: maximum penalties under national security law.

Data Theft

Identity Theft

Cyber Terrorism

Fraud



Data Protection Framework

- Data Collection: Lawful, transparent, and purpose-limited collection only
- Storage Rules: Encrypted storage with access controls and audit logs
- Data Sharing: Restricted third-party sharing with explicit user consent
- Retention & Deletion: Defined retention periods with mandatory deletion policies
- Breach Notification: Mandatory reporting to NCSA within 72 hours of detection





Corporate Responsibilities

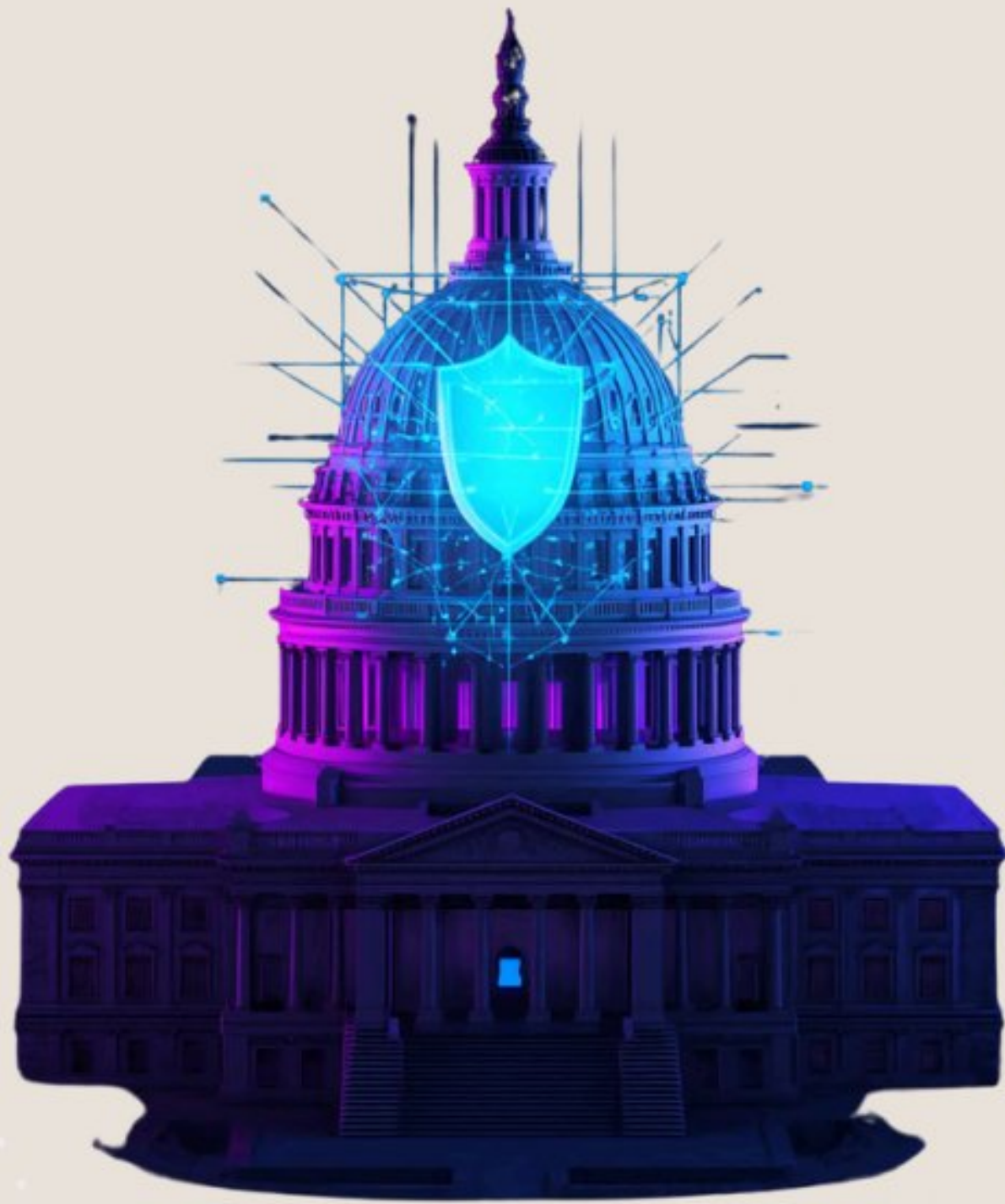
Organizations operating within VIT's must uphold strict cybersecurity obligations. These include enforcing data protection policies, implementing strong access controls, conducting regular security audits, and ensuring all employees receive cybersecurity training. Timely incident reporting to the NCSA is mandatory under the National Cybersecurity Law.

Data Protection

Password Policy

Employee Training

Security Audits



Government Responsibilities

The government plays a pivotal role in national cybersecurity governance. Key duties include protecting critical infrastructure, monitoring and investigating cyber threats, fostering international cooperation, and running public awareness programs to build a resilient digital society.

Infrastructure

Monitoring

Cooperation

Awareness



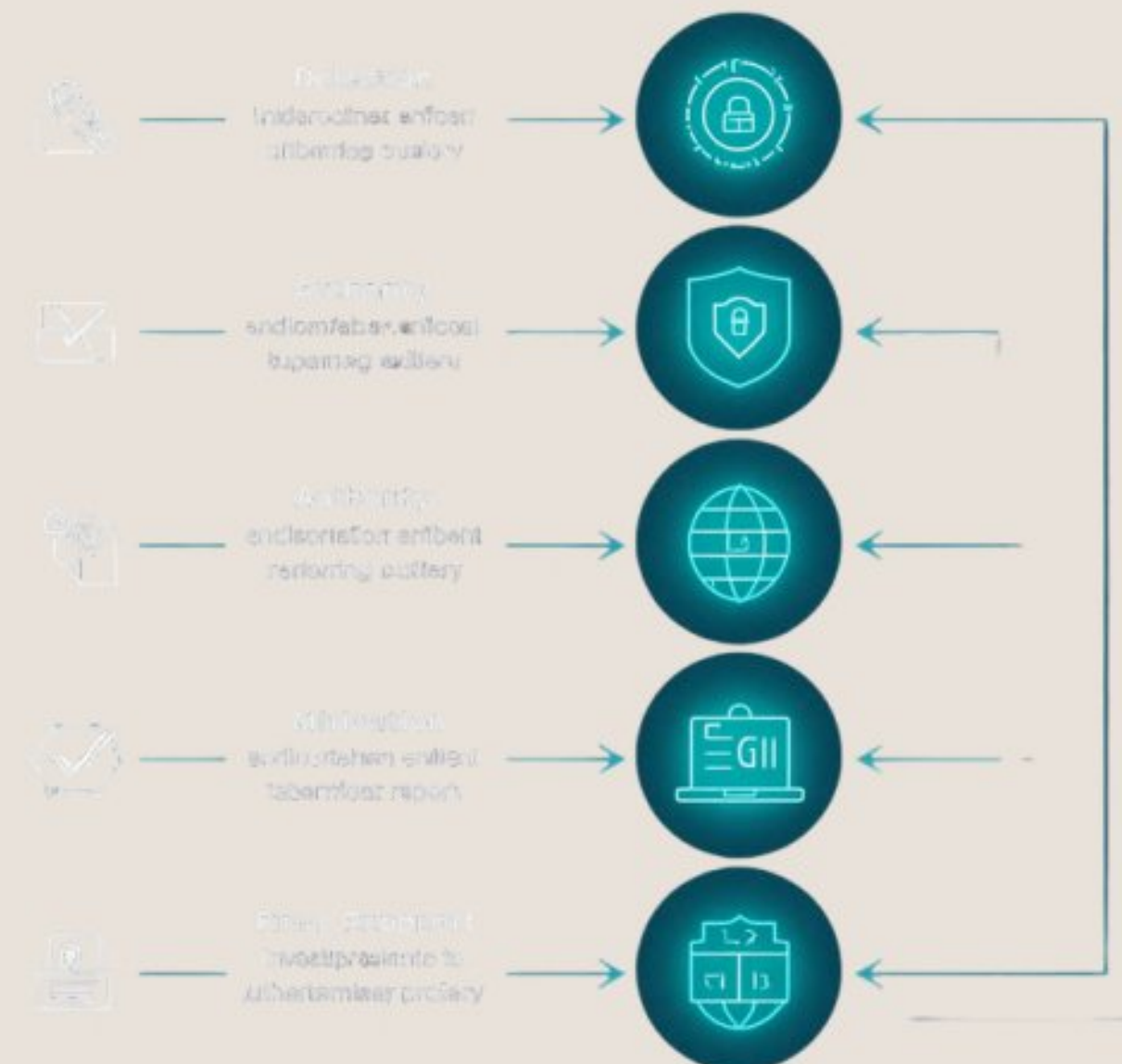
Cyber Incident Reporting

A structured incident reporting framework ensures swift response and recovery. Each stage — from initial detection to final report — is legally mandated under the National Cybersecurity Law to minimize damage and ensure accountability.

Detection → Investigation → Authority Reporting

Risk Assessment → Mitigation → Recovery → Final Report

Cybersecurity Incident Flow Reporting





National Cyber Security Authority (NCSA)

Organizational Structure: Director General, Cyber Intelligence Dept., Incident Response Unit, Legal & Compliance Division

Core Responsibilities: Monitor national cyber threats, coordinate incident response, enforce cybersecurity standards across sectors

Investigation Process: Threat detection → Evidence collection → Forensic analysis → Prosecution referral

Legal Enforcement Powers: Issue directives, conduct audits, impose penalties, and coordinate with law enforcement agencies



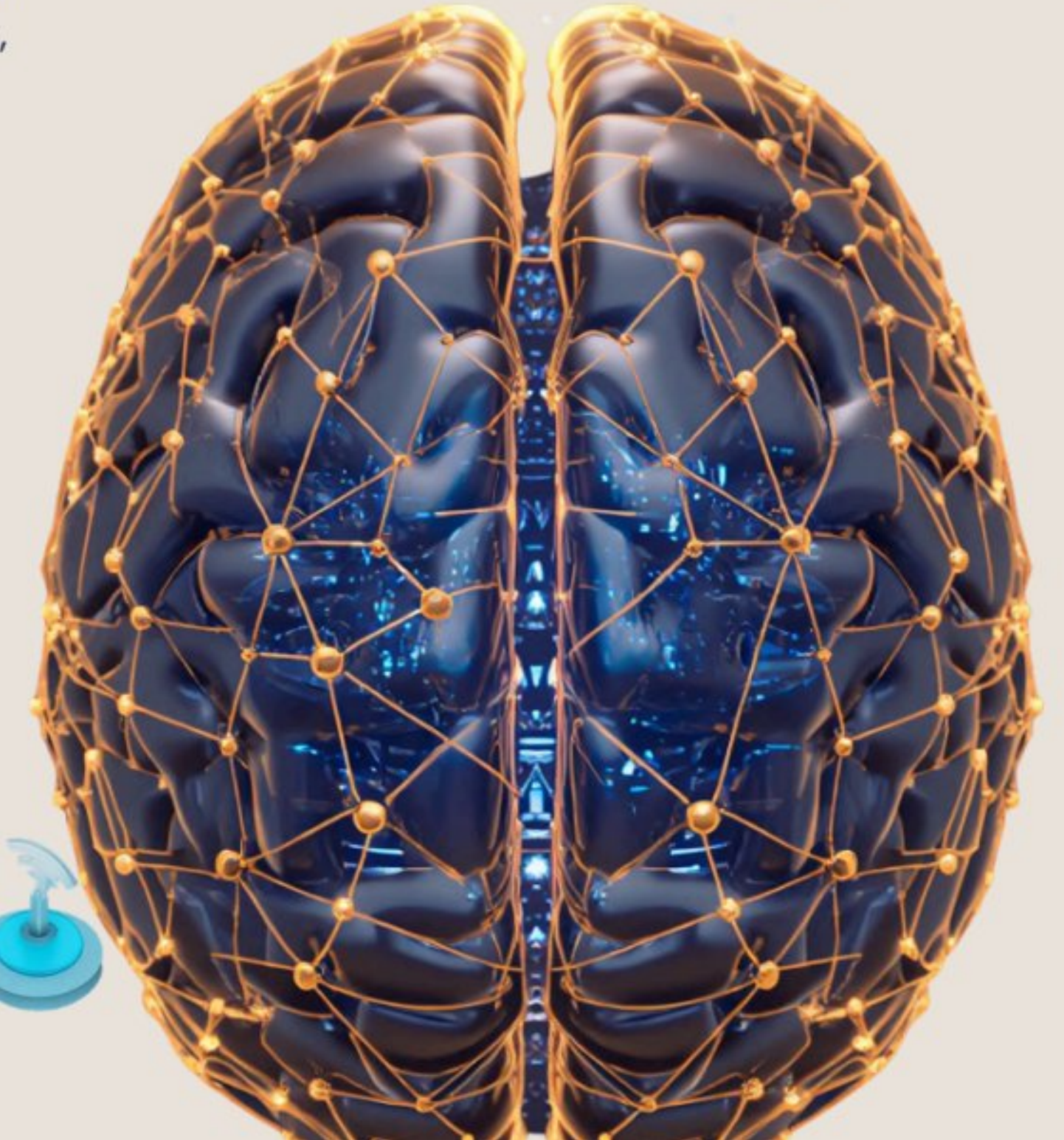
AI & Emerging Tech

The National Cybersecurity Law establishes regulatory frameworks for emerging technologies. AI systems, IoT devices, and digital identity platforms must comply with strict security, transparency, and accountability standards to protect citizens and national infrastructure.

AI & Deepfakes Regulation

IoT Devices & Cloud Computing

Digital Identity & Autonomous Systems





Critical Infrastructure Protection

Key national sectors require prioritized cybersecurity controls to safeguard essential services. These industries form the backbone of Cyberland's digital economy and public safety. The National Cybersecurity Law mandates strict protection standards, regular audits, and incident response protocols for each critical sector.



Banking

Healthcare

Aviation & Transport

Energy & Gov. Data



Awareness & Education

A national cybersecurity awareness strategy ensures all citizens, students, and professionals are equipped with the knowledge to identify threats, practice safe digital behavior, and contribute to a resilient cyber ecosystem across Cyberland.

Schools & Universities: Curriculum integration

Corporate Training & Social Media
Campaigns

Public Workshops & National Roadmap Rollout





International Cooperation

Cross-border collaboration is essential to combat cybercrime effectively. Nations must unite to investigate attacks, share intelligence, and coordinate enforcement. Joint operations and extradition frameworks strengthen the global cybersecurity ecosystem and deter transnational cyber threats.



Investigation

Info Sharing

Joint Ops

Extradition



Case Study 1: University Data Breach

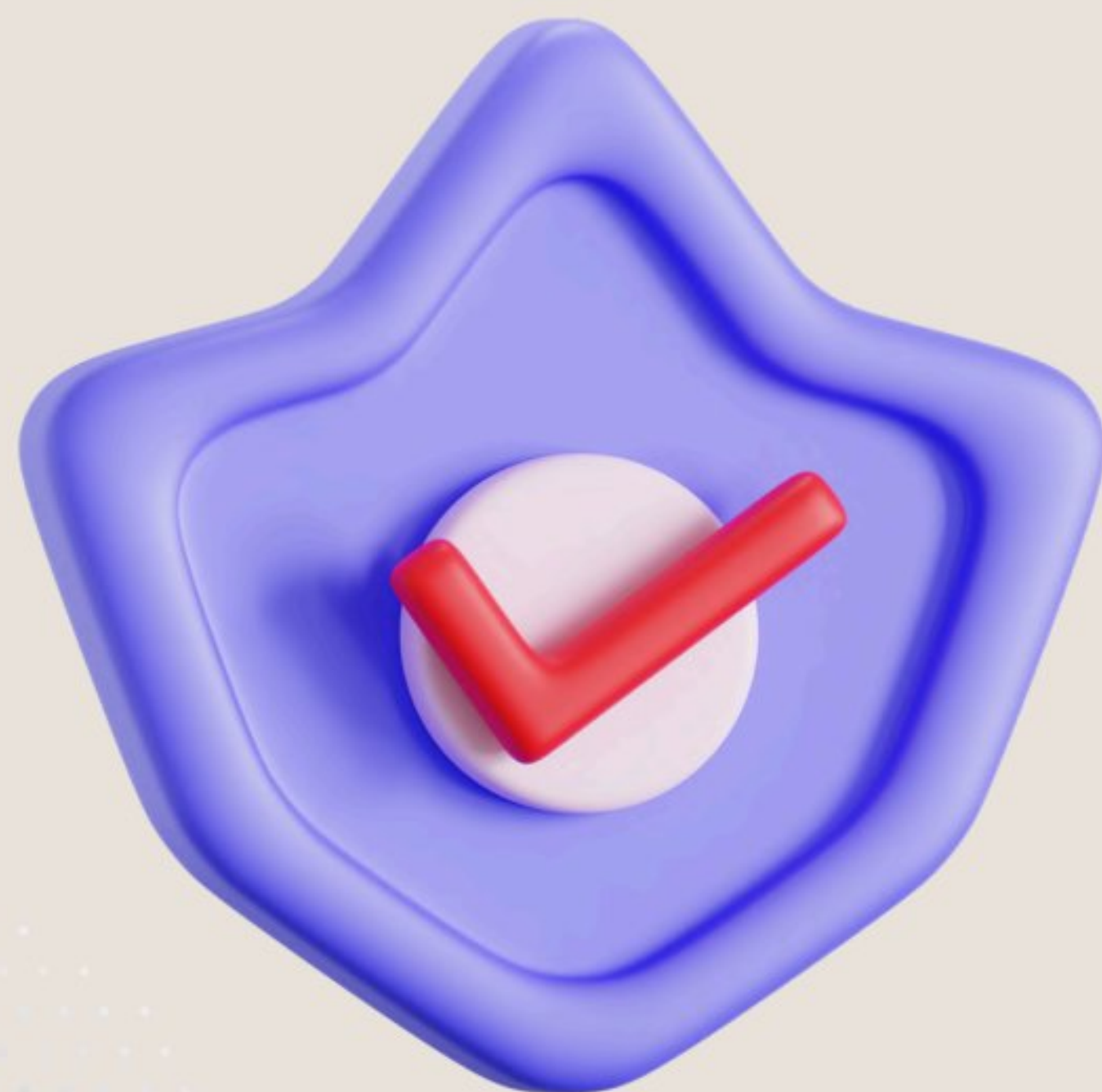
Incident: A major university suffered a breach exposing 50,000+ student and faculty records including personal data, financial details, and academic credentials.

Law Violated: Unauthorized access to systems, failure to implement adequate data protection measures, and delayed breach notification.

Investigation: NCSA traced the attack to a phishing campaign targeting faculty. Weak password policies and unpatched servers enabled lateral movement across the network.

Punishment: Institution fined; responsible IT officers faced disciplinary action; mandatory security overhaul ordered.

Lessons Learned: Enforce MFA, regular security audits, staff training, and timely breach notification protocols.



Case Study 2: Phishing & Financial Fraud

Incident: Attackers sent spoofed bank emails to university staff, harvesting credentials and siphoning ₹12L from institutional accounts.

Investigation: Digital forensics traced phishing kit to an overseas server; email headers and IP logs identified the perpetrators within 3 weeks.

Punishment: 5-year imprisonment + full financial restitution under Cyberland Cybercrime Act §42 (Financial Fraud).

Lessons: Enable multi-factor authentication, conduct regular phishing simulation drills, and enforce strict email filtering policies.



Case Study 3: Ransomware Attack

- Incident: Colonial Pipeline (2021) — DarkSide ransomware encrypted critical fuel infrastructure, halting 45% of U.S. East Coast fuel supply for 6 days.
- ⚖️ Law Violated: Critical Infrastructure Protection Act; unauthorized system access & extortion.
- 🔍 Investigation: FBI traced Bitcoin ransom (~\$4.4M); \$2.3M recovered via blockchain tracking.
- 🚓 Punitive Action: DarkSide operators sanctioned; international arrest warrants issued.
- 💡 Lessons: Segment OT/IT networks; enforce offline backups; mandate incident reporting within 24 hours.



Future Recommendations (2025–2035)

- AI Security & Quantum Computing: Develop quantum-resistant encryption standards and AI-driven threat detection frameworks
- Enhanced Privacy Protection & Digital Identity: Strengthen biometric data laws, digital ID verification, and privacy-by-design mandates
- Cyber Warfare Preparedness: Establish national cyber defense units, offensive countermeasure protocols, and inter-agency war-gaming exercises
- Strategic Roadmap: Phase-wise implementation from 2025 to 2035 aligned with global cybersecurity benchmarks





Key Benefits

The National Cybersecurity Law for VIT's – Cyberland delivers measurable improvements across all digital stakeholders. It strengthens institutional security posture, safeguards citizen data rights, empowers businesses with compliance frameworks, and builds national digital resilience and public trust.

Cyber Posture

Data Protection

Business Security

Digital Trust

Thank You



Vedant Bandal & Team



National Cybersecurity Law for VIT's – Cyberland



Vision: A secure & trusted digital future for all



Questions & Answers – Open for Discussion

